STAFF REVIEW · H446 056 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. Tree node with no children.

2. A queue.

3. Left subtree, right subtree, root.

4. 7, 5, 15, 10. Finish the left subtree, then the right subtree, then visit the root.

I Do — source worked steps

1. Explain — PostOrder(5) waits for PostOrder(2), which prints 1 then 2. The right call prints 7; only then is 5 printed: 1,2,7,5. null returns without printing. Node supplies Key, Left and Right.

We Do — guided CHECK

1. Explain
Order 9,4,12,6. Visit by depth, using FIFO.

You Do — source mark schemes

1. 1 [1],6 [1],3 [1],14 [1],10 [1],8 [1].

2. 8,3,10,1,6,14 [4]; start [8] [1] and after 8 [3,10] [1].

3. Postorder visits children before parent [2] allowing safe subtree cleanup [1]; BFS explores layers [2] first discovery gives shortest unweighted edge count [1].

Exit Ticket — source answer

1. False. Breadth-first ordinarily uses a queue.